

Applying Exponent Laws

Teacher's Edition

Read This First

This lesson rebuilds Math Nation 3.5 (SE pp. 124–129). The book's guided instruction (the $(2/5)$ pattern table) is Notes I; its Fernanda/Junior collaboration is Notes III; its equivalence table and matching activity are bank questions 5 and 10. Every value on the book's pages checks.

The one item shape the book gets subtly wrong is whiteboard 5 here, where two options have the same VALUE ($64/49$) for different reasons. The book's own teacher note flags it; ours keeps the item because the discussion it starts is the point, and the teacher note says how to run it.

The B1G-M misconception for this benchmark — taking the opposite instead of the reciprocal, $(-2/3)^{-5}$ as $(2/3)^5$ — is warm-up 4, Notes II, whiteboard 2, 5 and 9, and bank question 11.

Standard and Targets

MA.8.NSO.1.3 Extend previous understanding of the laws of exponents to include integer exponents. Apply the laws of exponents to evaluate numerical expressions and generate equivalent numerical expressions, limited to integer exponents and rational-number bases, with procedural fluency.

Clarification Rational-number bases with integer exponents; fractions and decimals both appear, and a decimal base is converted to a fraction before it is flipped.

Boundary A whole-number base is a fraction over 1, so $4^{-3} = (1/4)^3$ — said on Notes II so that 3.04 and 3.05 are one rule, not two.

Learning target I can rewrite a rational base with a negative exponent using a positive exponent, and apply several laws to generate an equivalent expression.

Essential question How do you rewrite a rational base with a negative exponent as an equivalent expression with a positive exponent?

Building on 3.04 — the negative exponent law; 3.02 — power of a quotient; Grade 6 — reciprocals of fractions.

Working toward 3.06–3.07 — evaluating and testing equivalence with everything at once; Thread A — the same laws with variable bases.

Where this sits in the course Second MA.8.NSO.1.3 lesson. 3.06+07 uses everything from 3.01–3.05 in one expression.

Lesson at a Glance

Windy Hill periods run **53 minutes**. The whiteboard round takes the remainder after the taught blocks — **19 minutes** in this lesson — and IXL takes the last 5. Slide numbers below are read from the deck when this edition is built.

Segment	Slides	Time
Settle in; post the learning target	1	1 min
Warm-Up	2-3	5 min
Notes	4-6	11 min
Example 1	7-8	3 min
Your Turn	9-10	3 min
Example 2	11-12	3 min
Your Turn	13-14	3 min
Whiteboards	15-32	19 min
IXL	33	5 min
Total		53 min

Materials Whiteboards and markers. Calculators are allowed on everything; $(-7/8)^{-2}$ on a calculator returns 1.306..., which is $64/49$ — worth converting once so students trust both forms.

Where the lesson lives Notes II — 'flip the numbers, not the sign'. Whiteboard 9 asks for it in writing. If the round shows B on question 5 for the value reason ($64/49$ either way), the class understands more than the board shows; ask for the -3 version.

No partner or group work anywhere in this lesson. Every Math Nation exploration this lesson draws on was written for pairs; here the same tables are worked from the front and on individual boards.

Vocabulary

reciprocal of a fraction — swap numerator and denominator: the reciprocal of $\frac{2}{5}$ is $\frac{5}{2}$; the sign stays where it was

rational base with a negative exponent — $\left(\frac{a}{b}\right)^{-n} = \left(\frac{b}{a}\right)^n$ — flip the base, make the exponent positive

Warm-Up — spaced retrieval

Four questions, two minutes silent, then the reveal. No reteach longer than one sentence per question.

1. Find the value: 2^{-3} $\frac{1}{8}$

retrieves: yesterday — 3.04 — negative exponent law

2. Rewrite with a positive exponent: $\frac{1}{7^{-2}}$ 7^2
retrieves: earlier this week — 3.04 — the law's second form
3. Which is larger, 4^3 or 3^4 ? 3^4 **(81 > 64)**
retrieves: earlier this unit — 3.01 — evaluating powers
4. Write the reciprocal of $-\frac{2}{5}$ $-\frac{5}{2}$
retrieves: prior grade — the prerequisite this lesson needs — Grade 6 — reciprocals of fractions (MA.6.NSO.2): flip the fraction, keep the sign

Slide-by-Slide Teaching Notes

Slide 1. Applying Exponent Laws 1 min

- Post the learning target. Say the 'today' line and nothing else yet.

Slide 2. Warm-Up — On your own. Four minutes. No notes. 4 min

- Q4 is today's move: a negative fraction's reciprocal is negative. A board with $5/2$ (sign dropped) or $2/5$ (opposite of the opposite) is the error this lesson is built to catch — the B1G-M lists 'takes the opposite instead of the reciprocal' as the misconception for this benchmark.

Slide 3. Warm-Up — Answers. 1 min

- Reveal. Ask for the band each question came from only if time allows.

Slide 4. Notes — Copy the table. 4 min

- Each row is the previous row divided by $2/5$, which is multiplying by $5/2$. Going down from $(2/5)^0 = 1$: $1 \cdot 5/2 = 5/2$, then $25/4$, then $125/8$. Students see that $(2/5)^{-1}$ is $5/2$ — the reciprocal — before the rule is stated.
- **OFF THE SLIDE, YOURS TO SAY:** 'To divide by two fifths you multiply by five halves. That is why the base flips.'

Slide 5. Notes 4 min

- State the rule and then the reason: $(a/b)^{-n} = 1/(a/b)^n = 1 \div a^n/b^n = b^n/a^n = (b/a)^n$. Then the sign question: $(-2/3)^{-5}$ is $(-3/2)^5$ — the minus rides along with the fraction. A student who writes $(2/3)^5$ took the opposite AND the reciprocal, which is the named misconception.
- **OFF THE SLIDE, YOURS TO SAY:** 'Reciprocal flips top and bottom. It does not touch the sign.'

Slide 6. Notes 3 min

- Fernanda: quotient of powers first, then power of a power, then rewrite the decimal as a fraction and flip. Junior: power of a quotient first, then quotient. Both land on $(2/5)^2$. The decimal-to-fraction step is the one to watch: $2.5 = 5/2$, so $(2.5)^{-2} = (5/2)^{-2} = (2/5)^2$.

Slide 7. Example 1 — Flip, then evaluate. 1 min

- SPOKEN ONLY. What is the base? What does the negative exponent tell you to do to it? Do the numbers or the sign flip?

Slide 8. Example 1 — Worked. 2 min

- Flip: $(-8/7)^2$. Square: $64/49$. Even power, so positive. A board with $-64/49$ squared only the numerator's magnitude and kept a sign; a board with $49/64$ never flipped.

Slide 9. Your Turn — Same steps, your numbers. Boards up when done.

2 min

- Catches a student who flips the sign with the fraction (the answer here is negative, and a board with $+8/125$ dropped it) and one who never flipped $(-125/8)$. $-8/125$ is right.

Slide 10. Your Turn — Answer. 1 min

- Reveal; name what a wrong board most likely did (see the note above).

Slide 11. Example 2 — Two laws, then a flip. 1 min

- SPOKEN ONLY. Same base twice — which law joins them? What is the exponent after that, and what does its sign tell you?

Slide 12. Example 2 — Worked. 2 min

- Product of powers: $-2 + 4 = 2$. The exponent is already positive, so no flip. Value $25/64$. The trap is flipping anyway because a negative exponent appeared somewhere in the problem.

Slide 13. Your Turn — Same steps, your numbers. Boards up when done. 2 min

- Catches a student who adds to -1 and stops there (a negative exponent is not a finished answer), and one who writes $(5/4)^{10}$ (added $7 + 3$ but kept the minus). $(4/5)^{10}$ is right.

Slide 14. Your Turn — Answer. 1 min

- Reveal; name what a wrong board most likely did (see the note above).

Slide 15. Whiteboards · Question 1 of 9 19 min for the round

- The rule, plain. A board with $(2/3)^4$ dropped the minus without flipping; a board with $-(3/2)^4$ made the exponent's sign the value's sign.

Slide 16. Whiteboards · Question 1 of 9

- Reveal. Scan the back row first; question the blank boards before the wrong ones.

Slide 17. Whiteboards · Question 2 of 9

- Sign inside the denominator. A board with $(5/3)^9$ took the opposite as well as the reciprocal — the named misconception.

Slide 18. Whiteboards · Question 2 of 9

- Reveal. Scan the back row first; question the blank boards before the wrong ones.

Slide 19. Whiteboards · Question 3 of 9

- Flip then square. A board with $-81/64$ kept a sign through an even power; $64/81$ never flipped.

Slide 20. Whiteboards · Question 3 of 9

- Reveal. Scan the back row first; question the blank boards before the wrong ones.

Slide 21. Whiteboards · Question 4 of 9

- Bigger numbers, same move. A board that tried to evaluate wasted the time: rewrite was asked.

Slide 22. Whiteboards · Question 4 of 9

- Reveal. Scan the back row first; question the blank boards before the wrong ones.

Slide 23. Whiteboards · Question 5 of 9

- Diagnostic. Note that A and B have the SAME VALUE (64/49) because the power is even — a student who chooses B for that reason is right about the value and wrong about the law. Ask them to redo it with exponent -3 , where B would be wrong in value too.

Slide 24. Whiteboards · Question 5 of 9

- Reveal. Scan the back row first; question the blank boards before the wrong ones.

Slide 25. Whiteboards · Question 6 of 9

- Diagnostic. $-2 - (-4) = 2$. The B board is the interesting one: it did the arithmetic right and then flipped out of habit.

Slide 26. Whiteboards · Question 6 of 9

- Reveal. Scan the back row first; question the blank boards before the wrong ones.

Slide 27. Whiteboards · Question 7 of 9

- Two bases that are reciprocals of each other: flip the first and they match. A board with 1 multiplied $2/5$ by $5/2$ and forgot the exponent flipped the first factor.

Slide 28. Whiteboards · Question 7 of 9

- Reveal. Scan the back row first; question the blank boards before the wrong ones.

Slide 29. Whiteboards · Question 8 of 9

- Product then power of a power: $(5 - 2) \cdot 3 = 9$. A board with $(2/3)^{21}$ multiplied $5 \cdot -2 \cdot 3$ wrongly; $(2/3)^6$ added $3 + 3$.

Slide 30. Whiteboards · Question 8 of 9

- Reveal. Scan the back row first; question the blank boards before the wrong ones.

Slide 31. Whiteboards · Question 9 of 9

- Written. This is the misconception the state guide names for this benchmark, in the student's own words. Full credit: the correct rewrite and a sentence that separates reciprocal from opposite. Value, if anyone asks: $-243/32$.

Slide 32. Whiteboards · Question 9 of 9

- Reveal. Scan the back row first; question the blank boards before the wrong ones.

Slide 33. IXL — Last five minutes. 5 min

- IXL: whatever is not finished is tonight's practice; SmartScore 67.

The Whiteboard Round — what each board tells you

Nine questions, one at a time, each with its reveal. Boards down until the cue, all up together, scan the back row first, question the blank boards before the wrong ones. Question 9 is written work. Every wrong option on a multiple-choice question is a named error; the benchmark it comes from is cited.

#	Question	Answer	What a wrong board says
1	$\left(\frac{2}{3}\right)^{-4}$	$\left(\frac{3}{2}\right)^4$	$(2/3)^4$ — the minus was dropped without flipping [8.NSO.1.3 negative exponent law]; $-(3/2)^4$ — negative exponent read as a negative value [B1G-M misconception]
2	$\left(\frac{3}{-5}\right)^{-9}$	$\left(\frac{-5}{3}\right)^9$	$(5/3)^9$ — took the opposite as well as the reciprocal [B1G-M 8.NSO.1.3 misconception: opposite instead of reciprocal]
3	$\left(-\frac{8}{9}\right)^{-2}$	$\frac{81}{64}$	$-81/64$ — sign kept through an even power; $64/81$ — never flipped [8.NSO.1.3 negative exponent law]
4	$\left(\frac{21}{4}\right)^{-5}$	$\left(\frac{4}{21}\right)^5$	$(21/4)^5$ — minus dropped without flipping; $4/21^5$ — flipped but applied the exponent to the denominator only
5	Which expression is equivalent?	A — $(-8/7)^2$	(B) took the opposite as well as the reciprocal [B1G-M 8.NSO.1.3 misconception: opposite instead of reciprocal]; (C) negative exponent read as a negative value [B1G-M 8.NSO.1.3 misconception: $-b$ vs b^{-1}]; (D) dropped the minus on the exponent without flipping [negative exponent law not applied]
6	Which expression is equivalent?	A — $(5/8)^2$	(B) flipped a base whose exponent was already positive [negative exponent law applied where no negative exponent remained]; (C) added $-2 + (-4)$ instead of subtracting [quotient law: subtract the exponents]; (D) multiplied $-2 \cdot -4$ [confused with power of a power]
7	$\left(\frac{2}{5}\right)^{-1} \cdot \left(\frac{5}{2}\right)^1$	$\frac{25}{4}$	1 — treated $(2/5)^{-1}$ as $2/5$ and multiplied by its reciprocal [negative exponent law not applied]; $5/2$ — only one factor counted

#	Question	Answer	What a wrong board says
8	$\left[\left(\frac{2}{3}\right)^5 \cdot \left(\frac{2}{3}\right)^{-2}\right]^3$	$\left(\frac{2}{3}\right)^9$	$(2/3)^6$ — added 3 + 3 in the power-of-a-power step [confused with product of powers]; $(3/2)^9$ — flipped a positive exponent
9	Marco says $(-2/3)^{-5}$ can be rewritten as $(2/3)^5$ because 'two negatives make a positive.' Is he right?	No — $(-2/3)^{-5} = (-3/2)^5$	'Yes' — opposite taken as well as reciprocal [B1G-M 8.NSO.1.3 misconception: opposite instead of reciprocal]

Question Bank — variation and answers

Variation. Questions 1–4 are one move each and vary only where the sign sits: no sign, a sign on the fraction, a sign in the numerator, a bigger fraction with no sign. Question 5 puts four sign placements side by side. 6–9 add one law before the flip (product, quotient, power of a power, then a decimal that must become a fraction). The pattern breaks at 7, where the exponent comes out positive and there is nothing to flip, and at 10, where one of four candidate expressions is not equivalent. The Additional sheet mirrors each position with different numbers.

The bank and its Additional sheet are posted, never assigned; they feed the quizzes. Answers below are the same values the keys print — both come from one source.

Question Bank

- $\left(\frac{4}{3}\right)^3$
- $\left(-\frac{8}{7}\right)^2$
- $\left(\frac{5}{-4}\right)^4$
- $\left(\frac{7}{22}\right)^7$
- (a) D (b) B (c) A (d) C**
- $\left(-\frac{14}{17}\right)^{-6} = \left(-\frac{17}{14}\right)^6$
- $\left(\frac{5}{8}\right)^2$
- $\left(\frac{2}{3}\right)^9$
- $2.5^{-2} = \left(\frac{5}{2}\right)^{-2} = \left(\frac{2}{5}\right)^2$
- (a) Yes — $\left(\frac{1}{6}\right)^3 = 6^{-3}$, so this is the same expression; value 6^6 . (b) No — the quotient gives 6^2 , not 6^{-8} ; this is 6^{-4} . (c) Yes — $\left(\frac{1}{6}\right)^{-2} = 6^2$, and $6^2 \cdot 6^4 = 6^6$. (d) Yes — it is the value of the expression written as one power.**
- A A**
- (a) $n = -2$ (b) $n = -3$**

Question Bank — Additional

- $\left(-\frac{9}{4}\right)^5$

2. $\left(\frac{10}{9}\right)^6$
3. $(-2)^8$
4. $\left(\frac{7}{2}\right)^3$
5. **(a) C (b) D (c) B (d) A**
6. $\left(-\frac{5}{4}\right)^{-10} = \left(-\frac{4}{5}\right)^{10}$
7. $\left(\frac{3}{7}\right)^{-3} = \left(\frac{7}{3}\right)^3$
8. $\left(\frac{4}{5}\right)^{-6} \cdot \left(\frac{4}{5}\right)^8 = \left(\frac{4}{5}\right)^2$
9. $1.5^{-2} = \left(\frac{3}{2}\right)^{-2} = \left(\frac{2}{3}\right)^2$
10. **(a) Yes – both simplify to 4^6 . (b) No – this is 4^{-4} . (c) Yes – $4^2 \cdot 4^4 = 4^6$. (d) Yes.**
11. **A A**
12. **(a) $n = -2$ (b) $n = -3$**

What the audit against Math Nation found

- Math Nation 3.5 pages verified: the $(2/5)$ table, $(3/2)^4$, $(-5/3)^9$, $(-9/8)^2$, $(4/21)^5$, $(2/5)^2$, the 6^6 equivalence table, the matching arrows, $(2/3)^9$, $(2/-13)^3$, 5^8 , $(4/-5)^{10}$.
- Practice L5's header says 'questions 6-10' over a table that is 7-10; not carried over.
- The book's warm-up (60^{63} vs 63^{60} , no calculation) is a fine number talk but retrieves nothing; the comparison 3^4 vs 4^3 takes its place in the warm-up's third band.
- Practice L5 and Homework L5 keys check; their matching items are bank question 5 in both banks.

What changed from the book

- 3.5.1 number-talk warm-up → four retrieval questions.
- 3.5.2 Guided Instruction → Notes I-II.
- 3.5.3 Collaboration (Fernanda/Junior) → Notes III.
- 3.5.4 Collaboration (equivalence table) → bank question 10.
- 3.5.5 Your Turn and 3.5.6 Ticket Out the Door → the round.